EXP 7: Logistic Regression (Classification)

CODE

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from collections import Counter
data = load_iris()
X, y = data.data[:, :2], data.target
model = LogisticRegression(max_iter=200)
model.fit(X, y)
pred = model.predict(X)
counts = Counter(pred)
plt.bar(["Setosa", "Versicolor", "Virginica"],
        [counts[i] for i in range(3)])
plt.xlabel("Iris Classes")
plt.ylabel("Number of Samples")
plt.title("Logistic Regression - Bar Graph")
plt.show()
print("Accuracy:", round(model.score(X, y)*100, 2), "%")
```

OUTPUT

